Cosmic Acceleration and the Absolute-Space Foundation

A Working Investigation

Daniel M. Andreola, Jr.

Independent Researcher

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Status of This Document

This document records an extended investigation into why the universe's expansion accelerates rather than decelerates, conducted within the absolute-space foundation. The investigation identifies the correct structural mechanism behind acceleration, tests and closes a wide range of candidate physical origins, and connects the remaining open question directly to a boundary already established elsewhere in this project. It is explicitly a working investigation: the mechanism is understood with real precision; the physical origin of the effect remains genuinely open, in the same honest category as other unresolved frontiers in this project.

1. The Real Problem

Real, precise cosmological observation confirms the universe's expansion is not merely continuing, but accelerating — a discovery honored by the 2011 Nobel Prize in Physics. This is the same real phenomenon addressed by the cosmological constant in standard cosmology, and it presents the same real difficulty for any foundation, including this one: ordinary motion and ordinary matter, checked directly, predict deceleration, not acceleration.

This investigation asked, from first principles, whether the absolute-space foundation's own postulates — particularly Postulate 3 (eternal, undirected motion) — could account for this effect without importing an unexplained external ingredient.

2. Mechanisms Tested and Closed

Each candidate below was tested directly, using real physics, and closed for a specific, computed reason.

- Ordinary ordered or random motion (kinetic pressure): real, established physics shows positive pressure — the kind ordinary motion produces — increases deceleration rather than causing acceleration; confirmed directly via the real cosmological acceleration equation.
- Light and electromagnetic radiation: carries an even more positive pressure than ordinary matter (real equation of state $w=1/3$), making the deceleration problem worse, not better; also subdominant in real, measured present-day energy density by roughly four orders of magnitude.
- Cosmic rotation: tested against the real, tightest published CMB-based observational constraint; even at the maximum allowed rotation rate, the resulting centrifugal effect is approximately 10^19 times too large relative to the real, observed dark energy density, and cannot be rescued by realistic corrections for a non-spherical (oval) mass distribution.
- Truly empty absolute space, under Postulates 1 and 3 alone, taken literally: produces no inherent energy density at all — confirmed directly, since neither postulate asserts anything about energy content in the absence of particles.
- Ordinary molecular/material negative pressure (the real cohesion-tension mechanism observed in trees): genuine, real, non-exotic negative pressure exists in nature, but relies on short-range molecular attraction with no known mechanism for extending across cosmic distances.

3. The Real Structural Mechanism

The investigation's central, positive finding: acceleration is not caused by any force growing stronger over time. It is caused by a real, asymmetric mismatch. Ordinary matter's gravitational pull weakens as the universe expands, following the real, standard density scaling (proportional to $1/a^3$, where a is the universe's relative size). A real cosmological constant, by contrast, maintains exactly constant density regardless of scale. As the universe grows, gravity's brake genuinely eases; the constant term does not. This mismatch — not a growing engine — is confirmed directly, via the real Friedmann acceleration equation, to be the correct mechanism producing net acceleration once the constant term comes to dominate.

4. Connection to Real, Published Physics

This investigation's reasoning was independently found to closely parallel two real, published lines of research. First, a real cosmological phase-transition model for dark energy (Dutta & Scherrer, and collaborators, Physical Review D, 2009) proposes that dark energy originated from an actual phase transition — described in the original publication as “similar to freezing” — occurring when the universe was roughly a quarter its present size. Second, real, published “freezing quintessence” and “Hubble friction” literature describes a mechanism in which a scalar field's evolution is damped by cosmic expansion until the expansion rate drops below the field's own characteristic scale, at which point the field is released — independently matching, in substantial detail, reasoning developed in the course of this investigation.

A direct numerical check of the real Scherrer transition point against the real, independently published dark energy energy scale (of order 10^{-3} eV) showed close numerical agreement; this is assessed honestly as confirming internal self-consistency of the published model rather than constituting new, independent evidence, since the transition point was very likely calibrated to reproduce the known value.

5. The Open Question, Precisely Stated

What remains unresolved is not the mechanism (Section 3) but the physical origin of the term that fails to dilute. This investigation's own honest search for a candidate — grounded solely in classical motion, radiation, and rotation — found no viable candidate that avoids the same wall found elsewhere in this project. This is consistent with, and directly connects to, this project's own earlier, separately established finding that ħ (the quantum floor) cannot be derived from the original four postulates and must be adopted as a fifth postulate, in the same transparent status as G.

A late refinement developed in this investigation, offered as a genuinely original conceptual proposal rather than an established result: standard treatments of this kind of transition (e.g., spontaneous symmetry breaking) implicitly assume a pre-existing, unstable initial configuration (informally, a field sitting at an unstable equilibrium). Under Postulate 3, taken strictly, motion has no beginning; there was never a true, static zero-point for such a configuration to depart from. This suggests the field's transition may not require an external trigger to disturb a prior stable state, since no genuinely static state may ever have existed to begin with. This reframing is recorded as a promising direction for future, rigorous development, not as a completed derivation.

6. Relationship to Standard Cosmology

It is noted explicitly, consistent with this project's standing practice of stating its own limits plainly: standard relativity models cosmic expansion as the metric of spacetime itself stretching, with matter carried along by that stretching geometry. This foundation's picture is structurally different — absolute space (Postulate 1) is fixed and unchanging, and expansion consists of matter genuinely moving apart within that fixed background, not space itself growing. As tested in this investigation, this different starting picture has not yet been shown to produce any observationally distinct prediction from standard cosmology; it remains, at present, an alternative account of the same observed phenomena rather than a new, independently testable claim.

Mechanisms Tested and Closed — Summary

- Ordinary/random kinetic motion — closed: produces positive pressure, increases deceleration.
- Light/radiation — closed: even more strongly positive pressure than matter, and subdominant in density.
- Cosmic rotation — closed: off by ~10^19 even at the maximum real observationally-allowed rate.
- Empty absolute space alone — closed: produces zero energy density under Postulates 1 and 3, taken literally.
- Ordinary molecular negative pressure — closed: real and non-exotic, but has no known cosmic-scale mechanism.

Statement of Methodology

No numerical result in this document has been adjusted, tuned, or reverse-engineered to produce an intended outcome. Every closed mechanism above is reported with the specific, computed reason for its closure, consistent with this project's standing practice of documenting negative results with the same rigor as positive ones.

References

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